

SECOND SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY**ENGINEERING CHEMISTRY**
MODEL QUESTION PAPER

Time: 3 hours

Maximum Marks: 75

PART A**I. Answer all questions in one word or one sentence. Each question carries one mark.****(9 x 1 = 9 Marks)**

1	Name the element with highest ionization energy		
2	Give an example for a redox indicator		
3	Write two examples of nucleophiles		
4	Write down a chemical reaction which is a natural first order reaction.		
5	The electronegativity of elements from left to right in a period of the periodic table		
6	Write down the IUPAC name of $\text{CH}_3\text{-CH}_2\text{-CHO}$		
7	The half life period of first order reactionof initial concentration		
8	Name a test that is used for the identification of protein.		
9	Aldehydes and ketones can be distinguished byreaction		

PART B**II. Answer any eight questions from the following. Each question carries 3 marks****(8 x 3 = 24 Marks)**

1	Noble gases have zero electron affinity. Why?		
2	How the principle of solubility product is made of in the separation of cations?		
3	Define accuracy and precision of an analytical result		
4	Differentiate homolytic and heterolytic fission.		

5	Write down the preparation methods of benzene by Dow's process and Cumene process		
6	Explain the following terms (a) Zwitter ion (b) isoelectric point		
7	What is Friedel craft's alkylation reaction?		
8	Explain the terms (a) order (b) molarity of a chemical reaction		
9	Define the half life period of a chemical reaction.		
10	Write down the Arrhenius equation and explain the terms.		

PART C

Answer all questions. Each question carries seven marks

(6 x 7 = 42 Marks)

III	What are the common errors in quantitative analysis? Explain the methods for its elimination.		
	OR		
IV	Explain the principle and working of paper chromatographic technique.		
V	Discuss the stability and structures of (a) carbanions (b) carbocations and (c) free radicals?		
	OR		
VI	Which are the different types of organic reactions? Explain with one example each.		
VII	Distinguish between Cannizzaro reaction and aldol condensation with suitable examples.		
	OR		
VIII	Distinguish between Markovnikov's addition and Anti Markonikov's addition with suitable examples		
IX	Derive the relation between K _p and K _c		
	OR		
X	Derive the expression of rate constant of first order reactions and the expression of half life.		

XI	Write down the open chain and ring structure of glucose .Explain the acetylation and osazone formation of glucose		
	OR		
XII	Write down the reaction of ethanol with (a) Luca's reagent (b) PCl_3 (c) PCl_5 and (d) SOCl_2 -		
XIII	(a) Explain the effect of temperature on the rate of a chemical reaction. (b) Define the terms threshold enegy and activation energy.		
	OR		
XIV	Explain the application Lechaltlier principle employed for the manufacture of sulphuric acid using contact process.		

Scheme of Valuation
Engineering Chemistry II

Time : 3 Hour
Max.Marks : 75

PART A

I. Answer all questions in one word or one sentence. Each question carries 1 mark.

1. Helium
2. Methylene blue (Any one example)
3. Cl^- , CN^- (Any two example)
4. Radioactive disintegration
5. Increases
6. 1-propanal
7. Independent
8. Ninhydrin test
9. Tollen's test

PART B

II. Answer any eight questions from the following, Each question carries 3 marks.

1. Electron affinity reflects the ability of an atom to accept an electron. It is the energy change that occurs when an electron is added to a gaseous atom. Atoms with stronger effective nuclear charge have greater electron affinity.
Group VIII elements, noble gases, have electron affinities near zero, since each atom possesses a stable octet and will not accept an electron readily.
2. Solubility-product constants can be used to devise methods for separating ions in a solution by selective precipitation. Selective precipitation is used to form a solid with one of the ions in solution without disturbing the other ions. We can use this method to effectively separate all of the ions in a solution. The entire traditional qualitative-analysis scheme is based on the use of these equilibrium constants to determine the correct precipitating ions and the correct strategy. Cations are usually classified into six groups. Each group has a common reagent that can be used to separate them from the solution. Because cationic analysis is based on the solubility products of the ions, meaningful results can be obtained only if separation is performed in a specified sequence. This is due to the fact that some ions of one group may also react with the reagent of another group. For example, both Ba^{2+} and Sr^{2+} will react with the SO_4^{2-} ion to form a solid. Therefore, mathematical calculations should be done before choosing the SO_4^{2-} ion for selective precipitation in a solution that may contain both Ba^{2+} and Sr^{2+} .
3. Accuracy of measurement as... "closeness of the agreement between the result of a measurement and a true value."

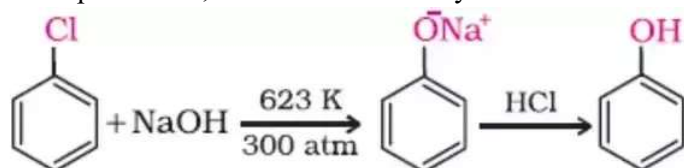
The term precision is used in describing the agreement of a set of results among themselves.

4. The symmetrical breaking of a covalent bond in which each departing atom takes one electron from the bonding pair is called homolytic fission

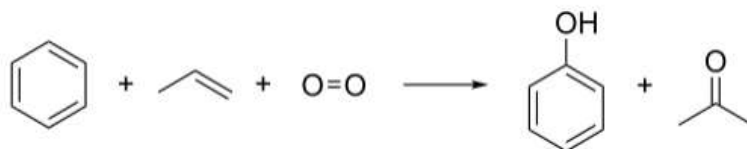
The unsymmetrical breaking of a covalent bond in which one of the departing atom retains the bonding electron pair is called heterolytic fission.

Homolytic fission results in the formation of free radicals while heterolytic fission results in the formation of carbocations or carbanions (Any two points)

5. Dow's process is one of the methods of preparation of phenol. In this method chlorobenzene is heated with aq. NaOH at about 623 K and 300 atm. pressure to form sodium phenoxide, which is acidified by dilute acid like HCl to form phenol.



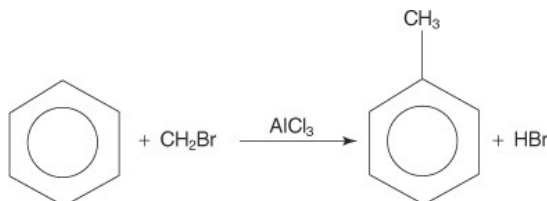
The cumene process is an industrial process for synthesizing phenol and acetone from benzene and propylene. This process converts two relatively cheap starting materials, benzene and propylene, into two more valuable ones, phenol and acetone. Other reactants required are oxygen from air and small amounts of a radical initiator.



6. A zwitterion is a molecule that has both positive and negative regions of charge.” In the solid state, amino acids exist as dipolar ions called zwitterions. Amino acids are the most common example of zwitterions. They are made up of an ammonium or amino group which contains a positive charge as well as a carboxyl group which contains a negative charge.

Main property of a Zwitterion is that it has an isoelectric point. This point is the pH value at which the charge in molecules is neutral.

7. An alkyl group can be added to a benzene molecule by an electrophile aromatic substitution reaction called the Friedel-Crafts alkylation reaction. One example is the addition of a methyl group to a benzene ring.



8.

Molecularity	Order
It is the number of reacting species undergoing simultaneous collision	It is the sum of the powers of the concentration terms in the rate law expression
It is a theoretical concept	It is experimentally determined
It has integral values only	It can have fractional values also
It cannot be zero	It can be zero
It provides no information on reaction mechanism	The slowest step in the reaction can be judged by the order of the reaction and this gives further information about the mechanism

9. The half-life of a chemical reaction can be defined as the time taken for the concentration of a given reactant to reach 50% of its initial concentration (i.e. the time taken for the reactant concentration to reach half of its initial value). It is denoted by the symbol ' $t_{1/2}$ ' and is usually expressed in seconds.
10. The Arrhenius equation is sometimes expressed as $k = Ae^{-E/RT}$ where k is the rate of chemical reaction, A is a constant depending on the chemicals involved, E is the activation energy, R is the universal gas constant, and T is the temperature.

PART C

Answer ALL questions. Each question carries 7 marks

III. Systematic or Determinate errors

Systematic errors can be determined, quantified and can be avoided. These comprise errors of a method, instrument errors and reagent errors. The following corrective actions can be applied to remove or eliminate such errors

1. Calibration of instruments and volumetric apparatus against reference standards and applying the correction factors
2. Use of high purity reagents and chemicals
3. Validation of method before use
4. Running blanks under identical conditions as for sample analysis and applying correction factor
5. Comparison of results through independent methods or interlaboratory comparisons
6. Using corrective methods such as standard addition or internal standards

7. Increasing analyte concentration prior to an analysis or derivatization with specific reagents (Any 5 points)

Random or indeterminate errors

Random errors manifest due to variations in a number of observations made by same observer with greatest care under as nearly identical conditions as possible. Generally such errors are not errors of the technique but personal errors of the observer.

Common personal errors are of the following types.

1. Incomplete drying of sample before weighing
2. Material loss during transfer of precipitates
3. Errors in transfer of solutions
4. Parallax errors in reading the rates and pipettes
5. Errors in making dilutions
6. Errors in observation of colour change during titrations
7. Transcription errors (Any 5 points)

OR

IV. Paper chromatography is a form of liquid chromatography where the basic principle involved can be either partition chromatography or adsorption chromatography.

In paper chromatography separation of component is distributed between phases of liquid. Here, one phase of liquid is water that is held amidst the pores of filter paper and the other liquid is the mobile phase that travels along with the filter paper. Separation of the mixture is the result that is obtained from the differences in the affinities towards the water and mobile phase when travelling under capillary action between the pores of the filter paper.

Paper Chromatography procedure

Selection of the ideal type of development: Based on factors such as the complexity of the solvent, mixture, paper, etc. the development type is chosen. Mostly either Radial or Ascending type of paper chromatography is employed because of the easiness they offer while handling and performing which ultimately leads to obtaining the chromatogram faster within a shorter duration of time.

Selection of Filter paper: As per the pores' size and the sample quality.

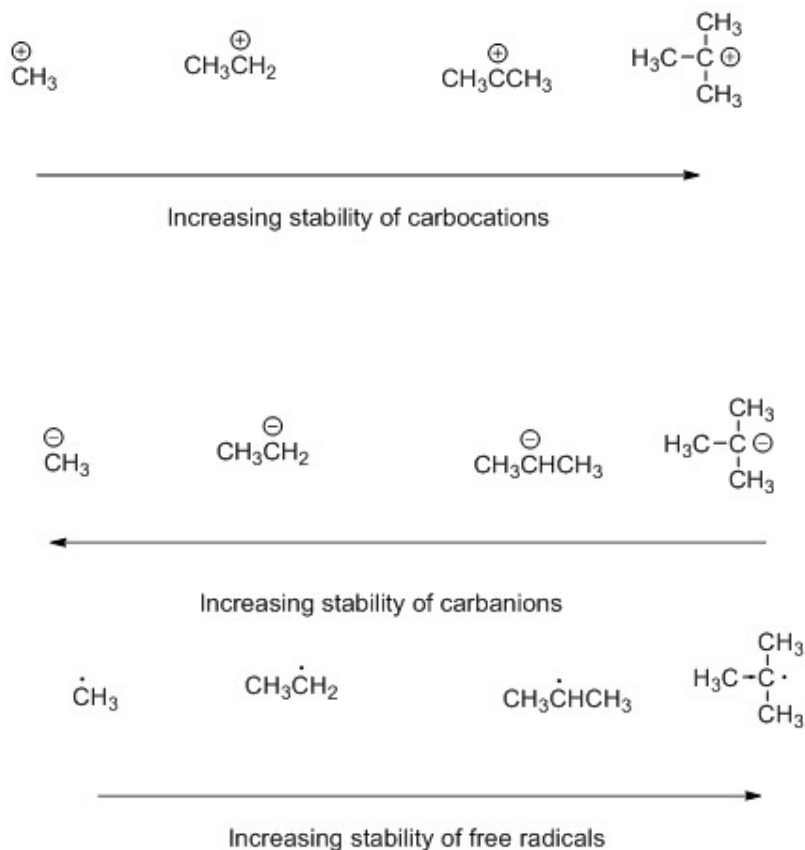
Sample preparation: This involves the dissolution of the sample in an ideal solvent that is being utilized in developing the mobile phase.

Sample loading or spotting on the paper: With the help of a capillary tube, micropipette, the sample is spotted on the paper at an accurate position. This promotes the interpretation of the chromatogram more quickly and easily.

Chromatogram development: This is carried by the paper immersion in the mobile phase. The mobile phase crosses over the sample on the paper because of the capillary action of the paper.

Drying of paper and detection of the compound: With the aid of air drier, the paper is dried as soon as the chromatogram is developed. On the chromatogram developed paper, the detecting solution is sprayed and dried thoroughly for the identification of the sample chromatogram spots.

V.



Stability order of carbocations increases as we move from primary to tertiary cation due to +I effect of methyl groups there is a redistribution of positive charge all over the molecule which reduces the intensity of positive charge on central carbon and increases the stability of the molecule.

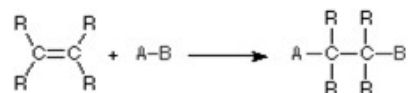
Stability order of carbanions decreases as we move from primary to tertiary anion because due to +I effect of methyl groups there is an increased intensity of negative charge on central carbon of tertiary carbanion which further makes it unstable.

Stability order of free radicals increases as we move from primary to tertiary free radicals, due to +I effect of methyl groups there is a redistribution of lack of electron all over the molecule which reduces the intensity of lack of electron on central carbon and increases the stability of the molecule.

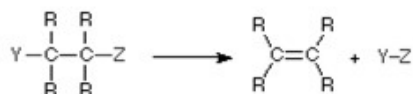
OR

VI. The four main reaction classes are additions, eliminations, substitutions, and rearrangements.

Addition Reaction



Elimination Reaction



Substitution Reaction



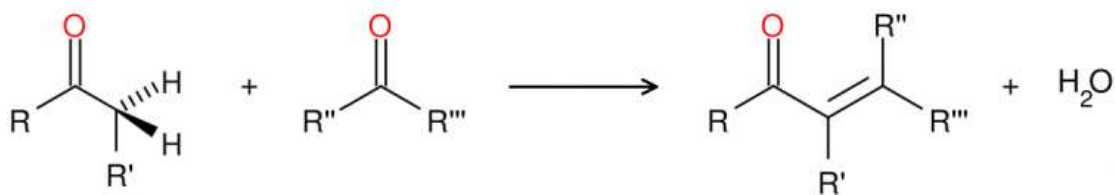
Rearrangement Reaction



In an addition reaction the number of σ -bonds in the substrate molecule increases, usually at the expense of one or more π -bonds. The reverse is true of elimination reactions, i.e. the number of σ -bonds in the substrate decreases, and new π -bonds are often formed. Substitution reactions, as the name implies, are characterized by replacement of an atom or group (Y) by another atom or group (Z). Aside from these groups, the number of bonds does not change. A rearrangement reaction generates an isomer, and again the number of bonds normally does not change.

(Any one example for each reaction)

VII Aldol condensation is an organic chemical reaction in which an enol or enolate ion reacts with a carbonyl compound, giving a conjugated enone. Furthermore, this reaction has two major steps: aldol reaction and condensation reaction. In the aldol reaction, an enol or an enolate ion reacts with a carbonyl compound to give a beta-hydroxyaldehyde or beta-hydroxyketone. Moreover, in the next step, dehydration occurs, giving a conjugated enone. The reaction is as follows:



Cannizzaro reaction is an organic redox reaction in which disproportionation of aldehydes give carboxylic acids and alcohols.

Cannizzaro reaction is an organic redox reaction in which disproportionation of aldehydes give carboxylic acids and alcohols. The reaction requires a strong base as the catalyst. In this redox reaction, a hydride transfers from one substrate to another. There, one aldehyde molecule undergoes oxidation, yielding an acid, while the other aldehyde molecule undergoes reduction, forming an alcohol. However, this reaction sometimes produces unwanted byproducts.

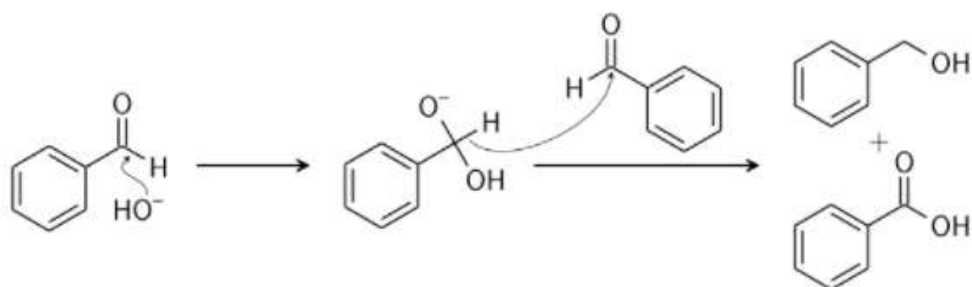


Figure 02: An Example for Cannizzaro Reaction

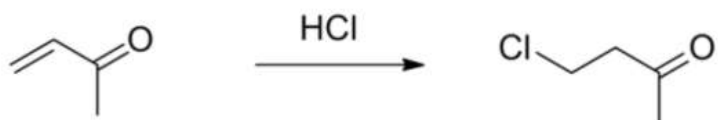
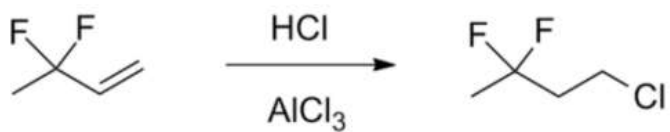
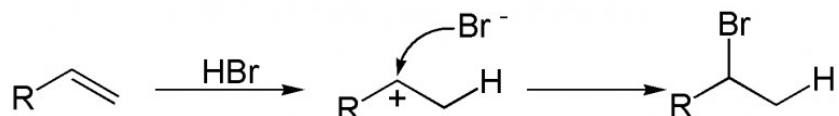
OR

VIII. The definition of Markovnikov rule is, when the addition of protic acid with the formula of HX (where X= halogen) or H₂O (considered as H-OH) to an alkene, hydrogen attaches to the double bonded carbon with the greater number of hydrogen atoms, while the halogen (X) attaches to the other carbon. Therefore, this rule is often interpreted as ‘the rich get richer’. The rule can be illustrated using the reaction of propene with hydrobromic acid (HBr) as follows.



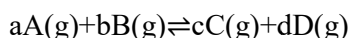
Figure 01: Markovnikov's Rule is illustrated by the reaction of Propene with Hydrobromic Acid

Anti-Markovnikov rule explains the opposite of the original statement of Markovnikov's rule. When HBr is added to an alkene in the presence of peroxide, H atom bonds to double-bonded carbon that has less C-H bonds, while Br bonds to the other carbon that has more C-H bonds. This effect is also known as Kharash effect or peroxide effect. The anti-Markovnikov addition also takes place when reactants are exposed to ultraviolet light. This is the exact opposite of Markovnikov rule. However, the anti-Markovnikov rule is not the exact reverse process of Markovnikov addition as the mechanisms of these two reactions are entirely different.



Anti-Markovnikov Rule (Any one example)

IX . Let the gaseous reaction is a state of equilibrium is-



Let p_A, p_B, p_C and p_D be the partial pressure of A, B, C and D respectively.

Therefore,

$$K_c = \frac{[\text{A}]^a [\text{B}]^b}{[\text{C}]^c [\text{D}]^d} \dots (1)$$

$$K_p = \frac{p_A^a p_B^b}{p_C^c p_D^d} \dots (2)$$

For an ideal gas-

$$PV = nRT$$

$$\Rightarrow P = \frac{n}{V} RT = CRT$$

Whereas C is the concentration.

Therefore,

$$p_A = [\text{A}]RT, p_B = [\text{B}]RT, p_C = [\text{C}]RT, p_D = [\text{D}]RT$$

Substituting the values in equation (2), we have

$$K_p = [A]^a (RT)^a [B]^b (RT)^b [C]^c (RT)^c [D]^d (RT)^d$$

$$\Rightarrow K_p = [A]^a [B]^b [C]^c [D]^d (RT)^{[a+b+c+d]}$$

$$\Rightarrow K_p = K_c (RT)^{\Delta n_g} \text{ (From (1))}$$

Here,

Δn_g = Total no. of moles of gaseous product – Total no. of moles of gaseous reactant

Hence the relation between K_p and K_c is-

$$K_p = K_c (RT)^{\Delta n_g}$$

OR

X. Any reaction is called a first order reaction if a change in concentration of just one reactant determines the rate of reaction. For a reaction as follows

$A \rightarrow \text{Product or products}$

$[A]_0$ = Initial concentration

$[A]_t$ = concentration at time t.

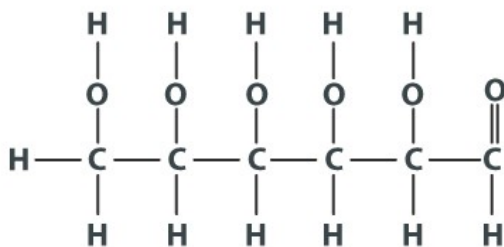
since reaction in first order-

$$\text{Rate} = -\frac{d[A]}{dt} = k[A] \Rightarrow \frac{d[A]}{[A]} = -k \cdot dt$$

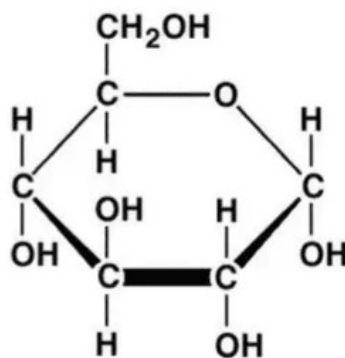
On integrating, we get

$$\int \frac{d[A]}{[A]} = -k \int_0^t dt \Rightarrow \ln[A] - \ln[A]_0 = -k[t - 0] \Rightarrow \ln \frac{[A]}{[A]_0} = -kt \Rightarrow \frac{[A]}{[A]_0} = e^{-kt} \Rightarrow [A] = [A]_0 e^{-kt}$$

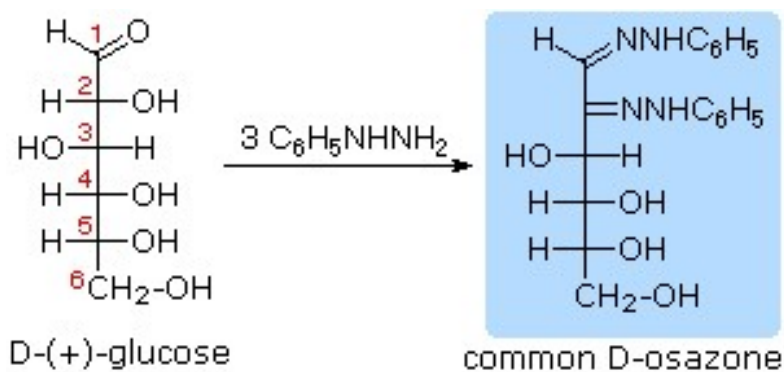
XI.



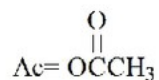
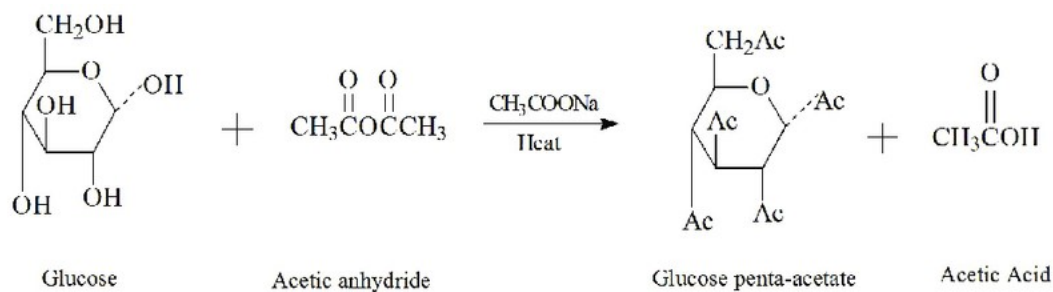
Open Chain Structure of a Glucose Molecule



Ring Structure of a Glucose Molecule



Osazone formation



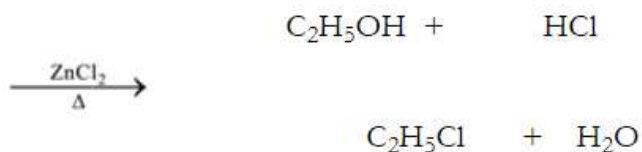
Acetylation of glucose

OR

XII.

(a) Luca's test

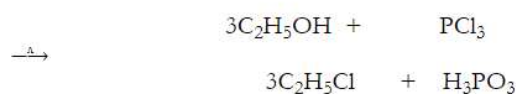
Example – 1: Preparation of ethyl chloride (Chloroethane) from ethyl alcohol (Ethanol):



Ethyl alcohol Conc. Hydrochloric acid Ethyl chloride

(b) PCl_3

Example – 1: Preparation of ethyl chloride (Chloroethane) from ethyl alcohol (Ethanol):

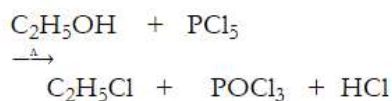


Ethyl alcohol Phosphorous trichloride ethyl chloride Phosphorus acid

Both the products are in the liquid state and are separated by fractional distillation.

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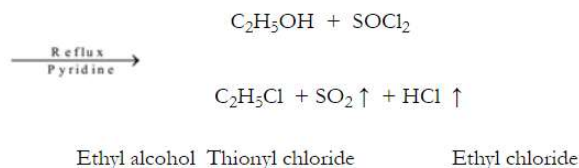
Example – 1: Preparation of ethyl chloride (Chloroethane) from ethyl alcohol (Ethanol):



Ethyl alcohol Phosphorous pentachloride Ethyl chloride
Phosphoryl chloride Hydrogen chloride

(d) SOCl_2

**Example – 1: Preparation of ethyl chloride
(Chloroethane) from ethyl alcohol (Ethanol):**



XIII.ARRHENIUS EQUATION - LAW

The rate constant of a reaction is a function of temperature. It is related to temperature by the Arrhenius equation.

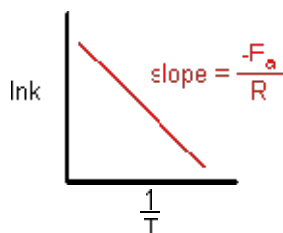
$$k = Ae^{-E_a/RT}$$

where k = rate constant

A = frequency factor or preexponential factor or Arrhenius constant.

E_a = activation energy of the reaction (J/mole)

ARRHENIUS EQUATION PLOTS - DIAGRAM



Applying log both sides and rearranging Arrhenius Equation we get,
 $\ln K = \ln A - E_a/RT$

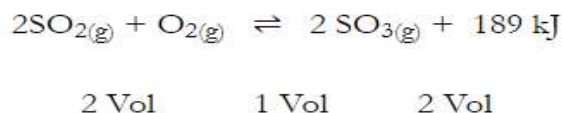
Activation energy: Energy given to the reactant to reach the threshold value of energy is called as activation energy.

Threshold energy: It is the minimum amount of energy at which reaction takes place.

OR

Application of Le-Chatelier's Principle to Contact process (Synthesis of Sulphur Trioxide):

H₂SO₄ is manufactured by contact process. In this reaction SO₂ is oxidized to SO₃. Sulphur trioxide is further used for manufacturing of sulphuric acid.



From this reaction it is clear that, the reaction is exothermic and accompanied by the decrease in volume.

Effect of Concentration:

By the law of mass action; increase in concentration of one of the reactant will shift equilibrium towards right. And here increase in concentration of sulphur dioxide in preference to oxygen has more effect.

Effect of Pressure:

Above reaction indicate that formation of sulphur trioxide takes place with decrease in volume. Hence increase in pressure will favour forward reaction. Optimum pressure for maximum yield of sulphur trioxide is about 1.5 atm to 1.7 atm

Effect of Temperature:

The reaction is exothermic, so lowering the temperature will favour forward reaction. But decrease in temperature results in the decrease in the rate of reaction. Hence temperature of 723 K is maintained and vanadium pentoxide is used as catalyst.